Assignment-5

Rishi Sharma

⇒ Section-A

A1. The purpose of a t-test is to determine if there is a significant difference b/w the means of 2 groups.

A2. The null hypothesis in a t-test states that there is no significant difference b/w the means of two groups.

A3. An F-test is typically used to compare the variances of two or more groups to determine if they differ significantly.

A4. ANOVA helps us test whether means of three or more groups are significantly different from each other.

A5. In a paired t-test, each observation in one group is paired with an observation in other group.

A6. The null hypothesis in paired sample t-test that the mean difference b/w paired observations is zero.

A7. The alternative hypothesis in paired samples t-test states mean difference b/w paired observations is zero.

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A8. Degree of Freedom = $n-1$

A9. To test the independence of attributes, the observed data are arranged in contingency table.

A10. F - Ratio = $\frac{\text{Between-group Variance}}{\text{within group Variance}}$

⇒ Section-B

B1. $\bar{x} = 158$ cm

$\mu_0 = 162.5$ cm.

$S^2 = 39.0625$ cm

$S = 6.25$ cm

$n = 10$.

Significance level (α) = 5% = 0.05.

$H_0 : \boxed{\mu > 162.5}$ & $H_a : \boxed{\mu < 162.5}$

↳ Null ↳ Alternative Hypothesis.

$$\Rightarrow t = \frac{\bar{x} - \mu_0}{S/\sqrt{n}} = \frac{158 - 162.5}{6.25/\sqrt{10}} \approx -2.27.$$

Decision: $t \approx -2.27 < -1.833$;

⇒ reject H_0

⇒ the data supports hypothesis that the average height is less than 162.5.

B2. Battery A : $n_1 = 10$, $\bar{x}_1 = 1000$, $S_1^2 = 100$

Battery B : $n_2 = 12$, $\bar{x}_2 = 2000$, $S_2^2 = 121$

Significance level (α) = 0.05.

⇒ Hypothesis - $H_0 : \mu_1 = \mu_2$

$H_a : \mu_1 \neq \mu_2$

⇒ $t = \dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{S_1^2}{n_1} + \dfrac{S_2^2}{n_2}}} = \dfrac{1000 - 2000}{\sqrt{\dfrac{100}{10} + \dfrac{121}{12}}}$

$\simeq -79.88$.

$t_{0.026} = \pm 2.262$

⇒ Decision : $|t| >> 2.262$ ∴ reject H_0

⇒ Conclusion - there is a significant difference b/w two battery means.

B3. $S_1 = 27.5$ $n_1 = 13$ $S_1^2 = 756.25$

$S_2 = 29.75$ $n_2 = 16$ $S_2^2 = 885.06$

⇒ Hypothesis :- $H_0 : \sigma_1^2 = \sigma_2^2$

$H_a : \sigma_1^2 \neq \sigma_2^2$

$\Rightarrow F = \dfrac{S_1^2}{S_2^2} = \dfrac{885.06}{756.25} \approx \underline{\underline{1.17}}$

$\Rightarrow$ critical value
for $df_1 = 15$ and $df_2 = 12$, $F_{0.01} = 3.07$

$\Rightarrow$ Decision
$F = 1.17 < 3.07$, fail to reject H_0.

$\Rightarrow$ Conclusion - The variances are not significantly different.

B4. Total Families = 2000, tea consumers = 1400
Hindu Families = 1800, tea consumers = 1236

$\Rightarrow$ expected values (E) -

$E_{Hindu+Tea} = \dfrac{1400}{2000} \times 1800 = 1260.$

$E_{nonhindu+Tea} = \dfrac{1400}{2000} \times 200 = 140.$

$x^2 = \dfrac{(O-E)^2}{E} = \dfrac{(1236-1260)^2}{1260} + \left(\dfrac{140-140}{140}\right)^2$

$\Rightarrow x^2 = 0.48.$

⇒ Critical value

for df = 1 and $\alpha = 0.05$; $\chi^2_{0.05} = 3.84$

⇒ Decision

$\chi^2 = 0.48 < 3.84$; fail to reject H_0.

⇒ There is no significant difference in heat consumption b/w Hindu and non-hindu.

B5. I : $n = 6$; $\bar{X}_1 = \frac{\Sigma x}{n} = \frac{64}{6} = 10.67$.

II : $n = 6$; $\bar{X}_2 = \frac{\Sigma x}{n} = \frac{35}{6} = 5.83$.

III : $n = 6$; $\bar{X}_3 = \frac{\Sigma x}{n} = \frac{55}{6} = 9.17$.

⇒ $SSB = \sum_{i=1}^{K=3} n_i (\bar{x})^2$

I : $6(10.67 - 8.06)^2 = 6(2.61)^2 = 40.89$.

II : $6(5.83 - 8.06)^2 = 6(-2.23)^2 = 29.87$

III : $6(9.17 - 8.06)^2 = 6(1.11)^2 = 7.39$

⇒ $SSB = 40.89 + 29.87 + 7.39 = \underline{\underline{78.15}}$.

$$\Rightarrow SSW = \sum_{n=1}^{k} \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2$$

I : 107.34

II : 18.84

III : 28.84

$\Rightarrow SSW = 107.34 + 18.84 + 28.84 = \underline{\underline{155.02}}$.

$\Rightarrow SST = SSB + SSW = 78.15 + 155.02$
$= \underline{\underline{233.17}}$.

$$\Rightarrow MSB = \frac{SSB}{df_{between}} = \frac{78.15}{K-1} = \frac{78.15}{2} = \underline{\underline{39.08}}.$$

$$\Rightarrow MSW = \frac{SSB}{df_{within}} = \frac{155.02}{n_{total} - K} = \frac{155.02}{18-3} = \underline{\underline{10.33}}.$$

$$\Rightarrow F\text{-ratio} = \frac{MSB}{MSW} = \frac{39.08}{10.33} = \underline{\underline{3.78}}.$$

$\Rightarrow$ Critical Value : $F_{0.05} \approx 3.68$ (From Table)

$\Rightarrow$ Decision -

$F = 3.78 > 3.68$, we reject H_0.

$\Rightarrow$ There is a significant difference in mean weights of rats fed with 3 different food types.

$\Rightarrow$ Section-C

9. Sample data : $\{50, 49, 44, 52, 45, 48, 46, 45, 49, 45\}$

Sample size $(n) = 10$

$\mu_0 = 50$ Kg

$\alpha = 5\%$ (0.05)

$$\Rightarrow \bar{x} = \frac{50+49+44+52+45+48+46+45+49+45}{10}$$

$$= \frac{473}{10} = \underline{47.3 \text{ Kg}}.$$

$$\Rightarrow s = \sqrt{\frac{\Sigma(x_i-\bar{x})^2}{n-1}} = \sqrt{\frac{62.40}{10-1}}$$

$$= \sqrt{6.933} \approx \underline{\underline{2.63 \text{ Kg}}}.$$

$$\Rightarrow t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}} = \frac{-2.7}{2.63/3.162} = \frac{-2.7}{0.831}$$

$$= \underline{\underline{-3.25}}.$$

$\Rightarrow$ $df = 10-1 = 9.$

$df = \pm 2.262$. (critical value)

$\Rightarrow$ Here $3.25 > 2.262$, reject H_0.

$\Rightarrow$ At 5% level of significance, there is enough evidence to conclude that the avg packing weight of the bags is not 50 Kg.

C2. First group: {18, 20, 36, 50, 49, 36, 34, 49, 41}

Second group: {29, 28, 26, 35, 30, 44, 46}

Level of significance (α) = 5% (0.05)

$$\Rightarrow \bar{x}_1 = \frac{18+20+36+50+49+36+34+49+41}{9} = \frac{333}{9} = \underline{\underline{37}}.$$

$$\Rightarrow \bar{x}_2 = \frac{29+28+26+35+30+44+46}{7} = \frac{238}{7} = \underline{\underline{34}}.$$

$$\Rightarrow s_1^2 = \frac{\Sigma(x_1 - x_i)^2}{n_1 - 1} = \frac{1134}{8} = \underline{\underline{141.75}}.$$

$$\Rightarrow s_2^2 = \frac{\Sigma(x_i - \bar{x}_2)^2}{n_2 - 1} = \frac{386}{6} = \underline{\underline{64.33}}.$$

$$\Rightarrow s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2} = \frac{1134+386}{14}$$

$$= \frac{1520}{14} = \underline{\underline{108.57}}.$$

$$\Rightarrow t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{S_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}$$

$$= \frac{37-34}{\sqrt{108.57\left(\frac{1}{9}+\frac{1}{7}\right)}}$$

$$= \frac{3}{\sqrt{108.57 \times 0.254}} = \frac{3}{\sqrt{27.56}}$$

$$= \frac{3}{5.25} \approx \underline{\underline{0.571}}.$$

$\Rightarrow df = n_1 + n_2 - 2 = 9 + 7 - 2$
$= \underline{\underline{14}}.$

$\Rightarrow$ critical value $= \pm 2.145$.

$\Rightarrow$ $0.571 < 2.145$, we fail to reject the null hypothesis, H_0.

$\Rightarrow$ At the 5% level of significance, there is no sufficient evidence to conclude that avg marks of two groups differ significantly.

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C3. Sample A : $\{66, 67, 75, 76, 82, 84, 88, 90, 92\}$

Sample B : $\{66, 74, 78\}$

Level of significance ($\alpha = 10\%$)

$$\Rightarrow \bar{x}_1 = \frac{66+67+75+76+82+84+88+90+92}{9}$$

$$= \frac{720}{9} = \underline{\underline{80}}.$$

$$\Rightarrow S_1^2 = \frac{\Sigma(x_i - \bar{x}_1)^2}{n_1 - 1} = \frac{734}{9-1} = \frac{734}{8}$$

$$= \underline{\underline{91.75}}.$$

$$\Rightarrow \bar{x}_2 = \frac{66+74+78}{3} = \frac{218}{3} \approx \underline{\underline{72.67}}.$$

$$\Rightarrow S_2^2 = \frac{74.67}{3-1} = \frac{74.67}{2} = \underline{\underline{37.33}}.$$

$$\Rightarrow F = \frac{S_1^2}{S_2^2} = \frac{91.75}{37.33} = \underline{\underline{2.46}}.$$

$\Rightarrow df_1 = 9 - 1 = 8$

$\Rightarrow df_2 = 3 - 1 = 2.$

$\Rightarrow$ critical value $F(0.05, 8, 2) = 19.41$ (upper)

$F(0.95, 8, 2) = 0.052$ (lower)

6/w

⇒ if F fails lower and upper critical values, we fail to reject Ho.

⇒ $0.052 < F < 19.41$ where $F = 2.46$.

⇒ At 10% level of significance, there is no sufficient evidence to conclude that variances of two populations are significantly different.

C4. $E_{ij} = \frac{(\text{Row total}) \cdot (\text{column total})}{\text{Grand Total}}$

$E_{11} = \frac{(800)(500)}{1000} = 400.$

$E_{12} = \frac{(800)(500)}{1000} = 400.$

$E_{21} = \frac{(200)(500)}{1000} = 100.$

$E_{22} = \frac{(200)(500)}{1000} = 100.$

⇒ $X^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$

(i) Low, Public :

$\frac{(370-400)^2}{400} = \frac{(-30)^2}{400} = \frac{900}{400} = 2.25.$

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(2) Low, Government:

$$\frac{(430-400)^2}{400} = \frac{900}{400} = \underline{\underline{2.25}}.$$

(3) High, Public:

$$\frac{(130-100)^2}{100} = \frac{(30)^2}{100} = \frac{900}{100} = \underline{\underline{9}}.$$

(4) High, Government:

$$\frac{(70-100)^2}{100} = \frac{(-30)^2}{100} = \frac{900}{100} = \underline{\underline{9}}.$$

$\Rightarrow$ X^2 critical value $= 3.841$.

$\Rightarrow$ $X^2 = 22.5 > 3.841$, reject H_0.

$\Rightarrow$ There is sufficient evidence to conclude that income level and type of schooling are not independent.

⇒ Section - D

D1. $\bar{x} = \dfrac{4.2 + 4.6 + 4.1 + 3.9 + 5.2 + 3.8 + 3.9 + 4.3 + 4.4 + 5.5}{10}$

$= \dfrac{43}{10} = \underline{4.3}$ (thousand hours).

$\Rightarrow S = \sqrt{\dfrac{\Sigma(x_i - \bar{x})^2}{n-1}} = \sqrt{\dfrac{3.22}{10-1}}$

$= \sqrt{\dfrac{3.22}{9}} = \sqrt{0.3578} \approx \underline{0.598}.$

$\Rightarrow t = \dfrac{\bar{x} - \mu_0}{S/\sqrt{n}} = \dfrac{4.3 - 4.0}{0.598/3.162}$

$= \dfrac{0.3}{0.1892} = \underline{1.585}.$

$\Rightarrow t_{critical} = \pm 2.622$

$\Rightarrow$ $1.585 < 2.262$, reject Ho.

⇒ At 5% level of significance, there is no sufficient evidence to conclude avg lifetime of electric bulbs differ from 4000 hours.

Q2. Sample A : $\{10, 6, 16, 17, 13, 12, 8, 14, 15, 9\}$

Sample B : $\{7, 13, 22, 15, 12, 14, 18, 8, 21, 23, 10, 17\}$

$$\Rightarrow \bar{x}_1 = \frac{10+6+16+17+13+12+8+14+15+9}{10}$$

$$= \frac{120}{10} = \underline{\underline{12}}$$

$$\Rightarrow S_1^2 = \frac{120}{10-1} = \frac{120}{9} = \underline{\underline{13.33}}.$$

$$\Rightarrow S_1 = \sqrt{13.33} = \underline{\underline{3.65}}.$$

$$\Rightarrow \bar{x}_2 = \frac{7+13+22+15+12+14+18+8+21+23+10+17}{12}$$

$$= \frac{180}{12} = \underline{\underline{15}}.$$

$$\Rightarrow S_2^2 = \frac{314}{12-1} = \frac{314}{11} = \underline{\underline{28.55}}.$$

$$S_2 = \sqrt{28.55} = \underline{\underline{5.34}}.$$

$$\Rightarrow t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{12 - 15}{\sqrt{\frac{13.33}{10} + \frac{28.55}{12}}}$$

$$= \frac{-3}{\sqrt{3.712}} = \frac{-3}{1.927} = \underline{\underline{-1.56}}.$$

$\Rightarrow$ critical value, $df \approx 19$
$\downarrow$
$t_{critical} = \pm 2.093$.

$\Rightarrow$ $1.56 < 2.093$, reject H_0.

$\Rightarrow$ At 5% level of significance, there is no sufficient evidence to conclude that diets A and B differ significantly in effect on weight increase.

Q3. $$\bar{x}_1 = \frac{8260 + 8130 + 8350 + 8070 + 8340}{5}$$

$$= \frac{41150}{5} = \underline{\underline{8230}}.$$

$$\Rightarrow S_1^2 = \frac{63000}{4} = \underline{\underline{15750}}.$$

$$\Rightarrow \bar{x}_2 = \frac{7950 + 7890 + 7900 + 8190 + 7920 + 7840}{6}$$

$$= \frac{47740}{6} \approx \underline{\underline{7956.67}}.$$

$$\Rightarrow S_2^2 = \frac{56572.22}{5} = \underline{\underline{11314.44}}.$$

$$\Rightarrow F = \frac{S_1^2}{S_2^2} = \frac{15750}{11314.44} = \underline{\underline{1.392}}.$$

$df_1 = 5 - 1 = 4$

$df_2 = 6 - 1 = 5.$

$\Rightarrow$ critical values are determined for $df_1 = 4$ and $df_2 = 5$

$F_{critical\ lower} = 0.104$ and $F_{critical\ upper} = 4.57.$

$\Rightarrow$ $F \doteq 1.392 \in [0.104, 4.057].$
Reject Ho.

$\Rightarrow$ At 5% significance level, there is no sufficient evidence to conclude that variances of two population are different.